WebGL virtual globe for efficient forest production planning in mountainous area

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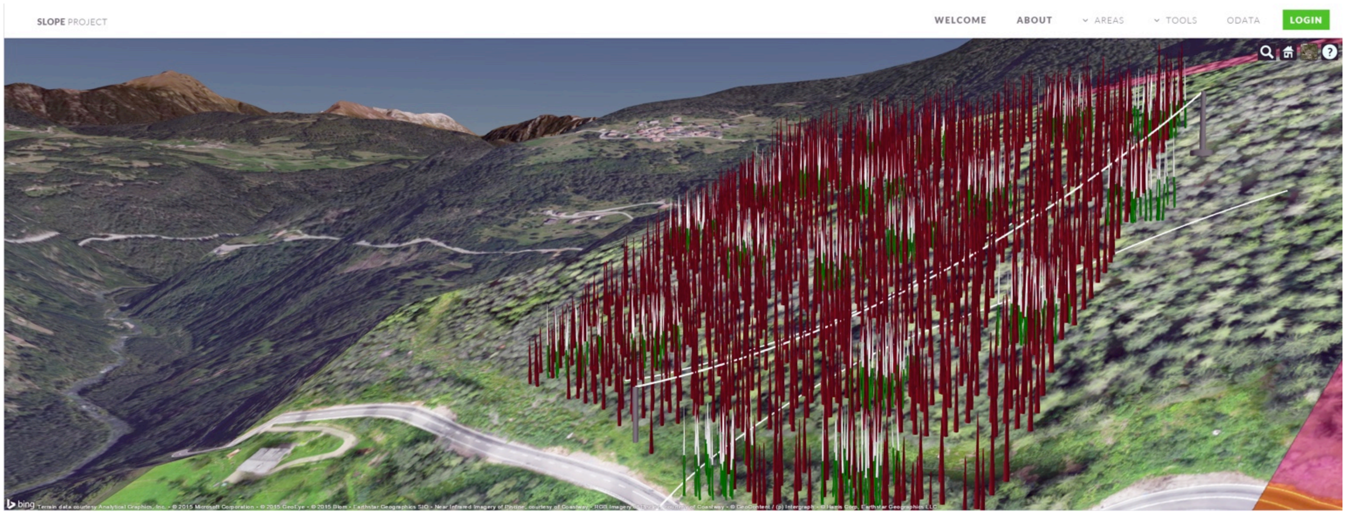

Figure 1: *The GeoBrowser WebGL 3D viewer showing the cableway simulator with low and high (red/green) detail tree stems reconstructed from unmanned aerial vehicle pictures and terrain laser scanner point cloud data.*

Abstract

This paper illustrates a 3D web-based visual simulation system that supports spatial-information based realistic modelling and real-time rendering of forest scenes. 3D visualization of forest landscapes will be used to understand stand ecological succession, landscape transformation helping regional planning and improving decision-making processes and forest management through what-if analysis tools.

The goal of the presented project is to deploy a web system able to easily provide access to forestry information, typically managed in GIS or dedicated system, ensuring the fruition of this information to non-geospatial experts like forest owners and field operators.Technologies such as HTML5, WebGL, CSS 3D and Canvas element have been chosen as the most suitable for the interactive visualization of the forest model placed in the context of a virtual globe representation of the Earth using a highly customized version of the CesiumJS virtual globe, called GeoBrowser3D. Powered by WebGL and state of the art software technologies it offers a three-dimensional Open Geospatial Consortium (OGC) compliant solution, integrated with computational and visual techniques for decision making environments.

This solution allows large scale GIS data visualization such as maps and forest models, as well as high detail small scale information such as point clouds or stems structures that are automatically generated according to inventory database and pre-designed template models. Additionally, forest activity planners can have access to larger scale datasets like roads and cadastral maps as well as single timber information.

The system has been tested and evaluated in applications of walkthrough simulation and forest visualization. The test site is situated in Northern Italy, close to Sover in the Trentino region.

CR Categories: I.3.7 [Three-Dimensional Graphics and Realism]: Virtual reality; I.3.8 [Applications]; H.3.5 [Online Information Services]: Web-based services;

Keywords: Virtual Globe, WebGL, Forest production, 3D Forest model, OGC.

1 Introduction

Forestry applications rely on geospatial data for managing forest resources. Recent development of UAV, optical systems acquiring multiple-view image data and fast photogrammetric processing algorithms have brought to the market new sources of 3D Digital Surface Models (DSMs) for the forest landscape, as well as terrestrial laser scanning (TLS) system supporting low cost, highly portable and rapid measurement of below-canopy 3D forest structure. These data, if integrated within an interactive and continuously updated model, can give information about detailed terrain topography, tree growth status, plots characteristics and

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other forest parameters; delivering a model-based planning system for forest mountain management.

Modelling and real-time rendering of forest scenes based on real world data is an important and challenging problem in computer graphics and geomatics [Bao et al. 2011]. In the last years, modelling and rendering of plants and forest scenes has been intensively explored in the field of computer graphics. However few researches have investigated the possibility to use 3D visualization techniques for representing real data of the forest.

This paper contains the first results of the European project SLOPE: Integrated processing and Control Systems for Sustainable Production in Farms and Forests. One of its aims is to develop a 3D web visual simulation system that supports geo-based realistic modelling and real-time rendering of forest scenes. The benefit of integrating visualization systems with modelling systems as part of a multi-disciplinary approach to forest-management planning and decision-making is one of the focal point of SLOPE.

The individual 3D tree profiles are generated from the terrestrial laser and UAV data. A DEM developed using available or collected data is the base for the rendered forest for the area of interest. The challenge of this work does not reside into extremely realistic forest rendering but to obtain a true-model of the forest using the real data collected from remote sensing and survey. This implies the generation of a 3D model that allows the planning and management of forest resources based on the real location and information about plot and trees.

3D visualization of forest landscapes will be used to understand stand succession, landscape transformation, and regional planning, and to improve decision-making processes and forest management in general. Furthermore combining 3D visualization of trees and ecosystem information with management practices permits the creation of realistic visual scenarios for forest management.

2 State of the art: virtual forest and 3D environment representation for forestry

Advances in computer data visualization in the past 30 years have opened new scenarios for natural resources studies and in particular forest environment. Scientific data visualization is continuously used in forestry planning activities giving a higher analytical power in critical decision in conditions where an accurate planning of resources in long term intervals is crucial. With the evolution of computer graphics the possibility to represent natural environment with photorealistic results has opened new perspective for forestry planners that were used to work with only photographic images and cartography maps. The validity of rendered forest with respect than real photographs was investigated since the 90s starting with forestry environmental planning, with prototypes of landscape visualization [Bergen et al., 1995] and ecological modelling simulation within a three dimensional model of the forest ecosystem [Zyda and Sheehan, 1997].

The great advantage, since the first forms of 3D visualization tools for forestry, is the possibility to manage planning problem from different scale levels: from large scale issues, as macro landscape design or fire prevention, to stand tree scale, where it is possible to investigate the nature of all the environment and finally to single tree perspective, where the forestry activities like timber harvesting, planting and thinning could be inspected [Wang et al. 2006]. In later years the evolution of remote sensing technologies with airborne and terrestrial LiDAR with rising resolution power, showed the new possibility of having a complete reconstruction from point cloud data of the real forest in a 3D model, opening up new scenarios for planners that now have the possibility to discover inaccessible area of land inside the reconstructed 3D model, but also introduced a new single tree approach in managing reconstructed data [Tao and Jian-hua, 2009]. In the optic of combining digital surface model of the forest area acquired from remote sensing surveys with 3D structural model of the canopy in the virtual environment, crucial became the accurate recreation of the trees [Guo and Niu, 2009]. In the first forest landscape models, they were often not represented in their geometrical complexity, but with simple primitives like cones. With the rising of graphical computational power the challenges in forest rendering became the achievement of realistic models with real time interactivity. The evolution of the tree representation from simple billboards impostors (represented by an X formed by two planar images) [Bao et al., 2009)] to a programmatic parametric geometrical approach for representing canopy and tree form [Candussi et al., 2005]. In later years researcher interested in creating more realistic and accurate representation of tree characterized landscapes investigated the possibility to create real time photo-realistic 3D forests, have studied a new model for lightings inside the forest [Bruneton and Neyret, 2012] and new representation of canopy reflectance inside steepness terrains [Fan et al. 2014].

Many techniques were studied to combine quality image and reasonable rendering performance during the following years. In 2006 Zhang et al. surveyed all the previous approaches for large scale rendering of forest environments and developed a new hierarchical representation for a forest model that permits a fast display (real-time interactivity) of rendering with great performance and good image quality including reasonable data storage cost. In earlier times other researchers concentrate on increasing realism and real-time performance: developed new technologies to create large (about ten thousand trees) realistic rendering of complex detailed tree structure with complex realistic representation of shadows without affecting the interactive experience of virtual environment exploration [Bao,et al. 2011].

The use of three dimensional forest models for planning and management support, has been studied in literature in different situations like resource estimation in virtual tropical forest, datasets analysis for carbon estimation [Brown, et al., 2005], as well as for directly forest trees management. Additional studies have been performed to measure forest indicators like plant growth through the deployment of an ecological model [Chen et al., 2005].

The use of virtual environment shows a new vision on decision support system based on GIS, moving the exocentric interface typical of two dimensional GIS to a new egocentric point of view. For management purpose it is essential to have the virtual landscape model correlated and linked to a central Forest Information System. In these way new types of interactions and monitoring are possible, like a remote controlling simulation of harvesting robot for virtual reality landscapes created with remote sensing data, where every single tree is mapped and references with a central forest database [Rossmann et al. 2009]. This research highlights the potentiality of four dimensional GIS for forest planning and the possibility of economic success for a forest system based on a trees database with cost estimation related to harvest simulation practice.

3 The virtual forest model

3.1 Input data

The generation of a reality based 3D virtual forest model requires development of sensor system, modelling and product-algorithms with capability to generate 3D structural ground truth in relation to the scale of acquisition. Typically, forest inventory database are organized in forest stands instead of individual trees. In this project, together with classical forest stand representation; tables contain also information on the single tree. The SLOPE project purpose is to represent and render interactive high quality dense forests tree by tree in real-time on major web browser.

The input data required for virtual forest generation include DTM for terrain representation, forest DSM obtained from UAV surveys, sample plots acquired by TLS and a forest inventory database.

A DTM is a regular grid of terrain elevation values on which each point is interpolated from spot height measurement or contour data. It is used to render the terrain and determine the height of the base of tree objects.

UAV sensor systems acquire multi-view optical image data, which can be used in photogrammetric processing to provide 3D data of forest. The system simultaneously produces accurate 3D models, i.e. Digital Surface Models (DSM) and spectral RGB image data shown in figure 1 and 2 respectively. The DTM and the new accurate DSM generation may very well constitute cost-efficient data sources to support forest management planning, providing the 3D height of the canopy.

High resolution data obtained through TLS and UAV survey (figure 4) pave the way to the capability to semi-automatically detect single trees in the forest, since these systems make available high resolution (0.1 m pixel size) DSM, and the scale of trees is much larger than the spatial resolution of the data.

At the end of the survey and data processing process there are four main input data for rendering the forest virtual model:

- The regular grids providing DTM and DSM;
- The point cloud acquired by TLS;
- The shape information about the trees surveyed by TSL, where each tree is represented by a series of circles at different height with the centre of the stem and the diameter;
- The shape information of the tree identified from the UAV data: a cone generated knowing the diameter on the basis and the height represents each tree.

The goal of SLOPE project is to deploy a web toolset able to easily provide access to this kind of information, typically managed in GIS or dedicated system to non-geospatial expert users like forest owners and operators with natural interaction paradigms like tree model selection.

The test site is situated in Northern Italy, close to Sover (Lat. 46°14'42.12"N, Lon. 11°19'54.00"E) in the Trentino region. The forest was surveyed in July 2014 to provide accurate single-tree ground truth data. In the area five field plots (15m radius each) were acquired by TLS and furthermore DBH, high and specie of each tree was accurately measured manually.

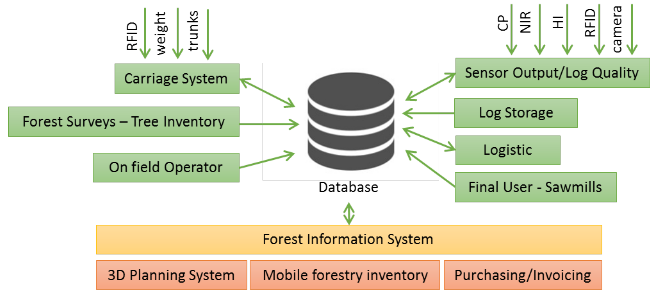

Figure 1: *Overall schema of the SLOPE system with at the center the Forest Information System (FIS).*

Optical image data were acquired in the same period using the UAV system, which consists of a standard and NIR cameras and an integrated navigation system for accurate positioning and attitude determination of the system. Georeferenced orthophotos and DSM with a pixel size of 0.10 m were produced.

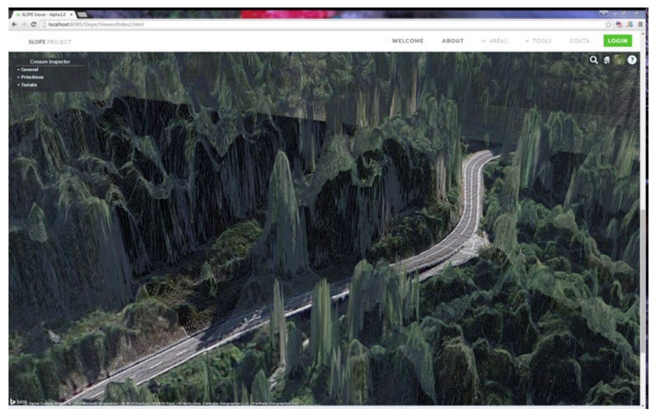

Figure 2: *Forest digital surface model (DSM) wireframe.*

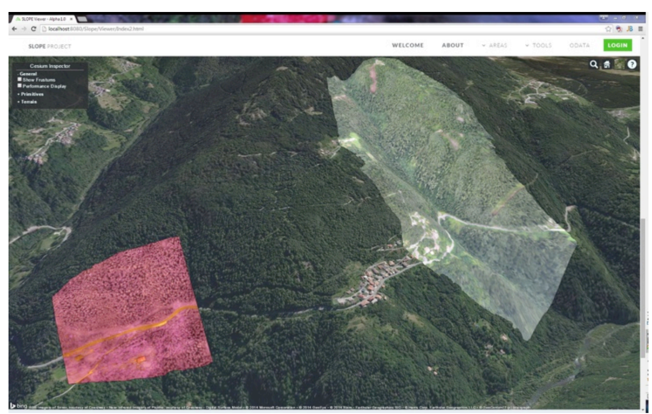

Figure 3: *UAV near infrared (NIR) and orthorectified imageries.*

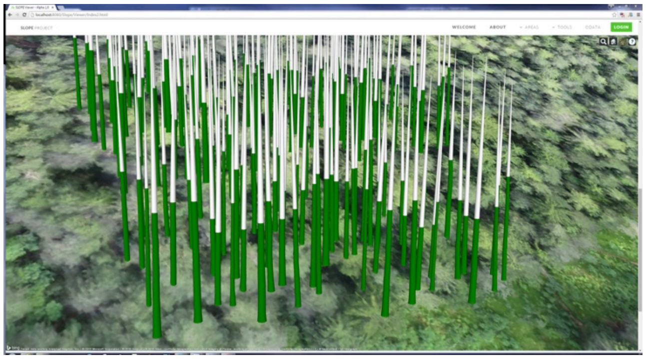

Figure 4: *Tree stems reconstructed from terrestrial laser scans (TLS)*.

3.2 Forest analysis integration and forest inventory in the three dimensional forest model

The four typology of input data retrieved during the survey of the pilot area are the basis for the development of the full 3D forest. By combining these kind of data it is possible to create a reproduction of a counterpart of the real forest area in a multiscale visualization. The virtual globe paradigm allows inspecting the survey area from different resolution and level of details, from the whole world to the single tree accuracy, giving also the possibility to combine the different datasets on the same area.

Thanks to this reconstruction work it is possible to provide a 3D management platform to examine the forest environment from the regional scale to the single tree. The first level of analysis examines the information at the whole forest scale using Earth Observation (EO) data and topographic maps. Satellites images gives a wider vision of the survey area, are used to find big patches of trees and in case of multi-spectral or hyper-spectral sensors, can be used to evaluate for instance the health state of the woods.

Analysis using multispectral satellite images (RedEye sensor) has been made on the test area. Temporal series of images have been processed in order to evaluate the characteristics of the plot through different indexes: Normalised Difference Red Edge (NDRE), which can be directly related to the level of Chlorophyll in the wood and Canopy Chlorophyl Content Index (CCCI) that can be used as an ancillary measurement to monitor also the Nitrogen and Clorophyll level.

The sets of maps generated from EO vegetation analysis are used as a coverage data for the area and can be utilized as a representation of the historical series within the geographical information system virtual globe giving to the forestry planner a wider scale dimension of the forest potentiality in the wide area of interest. To support planning and management of forest resources it is important to consider additional information like cadastral, logistic and environmental data. These data are typically made available as map layers containing the geographical related information. Nowadays many of this kind of datasets were made available as "Open-Data" making them freely available and giving the possibility to republish without restrictions. In this work these "Open data" have been exploited through a workflow that starts from raw data and arrives to a map service deployment, which will be shown by the 3D web-client.

Therefore, WebGL 3D forest system became the gateway [Craglia et. al 2012] to access through the web to remote sensing analysis and geographical information of the local mountain forests allowing owner and operators to optimise their activities.

The described system is a high level tool that allows a large scale planning of the forest resources; however mountain forest management needs a more accurate way to visualize data about vegetation within a smaller scale sufficiently detailed to represent up to a single tree timber. Understanding precisely the economic value of the selected harvesting parcel means optimize the tree felling to area which could give the higher profit preserving the silvicultural equilibrium. These requirements impose a shift on the overall system from data acquisition to visualization.

The first requirement is to identify each single tree on the plot, in order to achieve this result high-resolution spatial information. Have to be acquired. For instance, UAV images provide a more resolute smaller scale data of the forest. Starting from these georeferenced image sets it is possible to generate a point cloud densification of the area surveyed and a high-resolution DSM, in a form of raster geo-referenced image or elevation grid. The output DSM of the forest area can arrive to a resolution of approximately 20 cm and will be used within the planning tool, recreating the shape of the area and in particular of the digital canopy height model.

The DTM/DSM height-maps data and raster images are exposed respectively as a Tile Mapping Service (TMS) and through an OGC Compliant Web Mapping Service (WMS), which are completely interoperable with any other Geographic visualization service in the expanded Forest Information System scenario. A key aspect is that there are different level of detail needed in our virtual globe representation both for the orthophotos and the terrain height maps. In particular, for the height maps, the multiscale representation varies from a decimetres scale representation of the surface of the area of interest to the 10 meters scale resolution to represent area in the boundaries of the work site but not involved in planning analysis. In our system has been developed a Terrain Tile Map Service that expose tiled maps with different maximum level of detail, without oversampling low resolution terrain maps and supporting the visualization of all the level of details provided by the height map resolution. In this way within the planning tool it is possible visually explore a strict high resolution area where the terrain is represented with a centimetre scale detail and with the same zoom level the boundaries wider terrain areas without experience any sort of artifacts due to over sampling.

The resolution of the digital canopy model obtained by UAV imagery permits, through local maxima algorithm and watershed segmentation to detect single tree and retrieve accurately the number of the trees of that forest parcel [Jakubowski et al. 2013]. In this way, due to the geo-reference of the UAV imagery it is possible to reconstruct the geographical position of the single tree segmented by the aforementioned procedure, and give different identification code to the different tree retrieved: this data is correlated in the forest visualization model as in a forest inventory database.

To complete the digital reconstruction of the forest the information about stems are needed. The TLS processing can segment from the data the trunks information and size, thus giving an average value of the expected timber volume completed with information like trees number average diameter and also an the tree density, which is essential to maximize the timber production.

These forest parameters have been traditionally estimated on local models for each tree species (e.g. tariff method) which provide a relationship between the measured parameters (i.e. tree DBH, tree height and species) and the target parameters to estimate (i.e. tree volume) but these approaches are not always accurate. Through TLS it is possible to identify individual trees from the raw scanner point cloud, creating 3D models to accurately recreate an inventory plan of the forest stands.

Surveying different TLS samples within a forest it is possible to statistically infer the main parameters (i.e. tree DBH, and species) for the whole forest. With a combination of this information acquired from the ground with the information derived from UAV data, it is possible to define main geometrical parameters (DBH and height) for the whole forest. The trees are reconstructed through a geometry primitive, that reproduce the main morphological characteristics of the tree trunks (height, diameter at different height and quality index) and permit a visual exploration of the plot, to let the user inspect information for each single tree within the three dimensional virtual environment. The conical representation of the tree is functional for the forest planner for a rapid analytics of the economic value of the single stem and augments the decisional support power of the visual virtual environment representation of the forest plot. In this way, the final achievement is represented by a dataset where every tree is mapped inside a data structure representing the whole plot and the parameters for each single tree.

Finally, using this 3D forest tool the planner of the forestry activity has access to both larger scale data, like roads and cadastral datasets, and single timber information. In this way an efficient harvesting planning system to manage activities like logistic harvesting parcels decision and other common functionalities typical of the GIS as well as smaller scale planning aspect like single timber analysis and detailed forestry work managing can be setup into a usable web toolset.

4. Implementation

4.1 Visualization of forest on a WebGL based virtual globe

In order to achieve a good end-user experience with the huge amount of sensor data at our disposal, it is essential to provide a smooth and reactive visualization of the forest content with the highest detail possible. This is only achievable by leveraging the built-in hardware capabilities of modern desktop and mobile computers with a specific focus on discrete graphic cards. The so called hardware acceleration can be exploited with the adoption of specific standard graphics libraries defining a set of functionalities natively supported by the graphics cards. One of these is OpenGL (OpenGL, JOGL, etc.), a cross-language, multiplatform application programming interface (API) for 2D and 3D graphics developed by Silicon Graphics Inc. and widely supported by desktop and mobile hardware (i.e. smartphones). OpenGL appears in several different contexts going from mobile (OpenGL ES) to web (WebGL) and desktop. One of the major technologies which have spread in the last years together with the evolution of HTML is the WebGL technology which is becoming extensively supported by all the major browsers.

In the last decade, HTML evolved from standard static pages, introducing rich interaction by allowing portions of a page to be changed dynamically with AJAX techniques. The ways in which pages could be changed with Ajax were constrained by the graphical features of HTML and CSS but in the last few years several browser were developed with the new features brought by HTML5. With HTML5 and its collection of technologies, the web browser has become a platform capable of running sophisticated applications that rival native code in features and performance and a great part of its innovations are centred on advanced graphic techniques such as WebGL, CSS3 3D and the canvas element and its 2D drawing context API.

Each of these features has its strengths, weaknesses and technical trade-offs, and each has a role to play in delivering interactive and visually compelling 2D/3D experiences. HTML5 brings rich graphics to the web thanks to the essential browser improvements performed in the last years regarding Javascript virtual machines, accelerated compositing, animation, multi-threaded programming (Web Workers), full duplex TCP/IP networking (WebSockets), local data storage, and more that developers can use to deliver world-class application functionality. These features, taken together with WebGL, CSS 3D and the Canvas, represent a complete platform for delivering connected visual applications on any computer or device.

For these reasons, the aforementioned technology has been chosen as the most suitable for the interactive visualization of the forest model placed in the context of a virtual globe representation of the Earth. The adoption of a virtual globe of the entire Earth allows large scale GIS data visualization such as maps and tree models, as well as high detailed small scale information such as point clouds at the same time. Additionally, the coverage of the entire Earth surface allowing the parallel visualization of different datasets in different places simplifies the planning and monitoring of wood production.

Among the several WebGL 3D globe alternatives available on the web a highly customized version of the CesiumJS virtual globe, called GeoBrowser3D has been developed. Powered by WebGL graphics and state of the art software technologies, GeoBrowser3D offers a three-dimensional OGC compliant solution, integrated with computational and visual techniques for an ideal decision making environment. Its main features are the following:

- Advance data management of multi-dimensional GIS and satellite data from different service providers;
- Visualization and management of raster information from web mapping and tiling services and vector cartographic data (Shapefiles, WFS, WFS-T);
- Loading and visualization of terrain data from DSM and DTM at multiple resolution;
- Real-time data support using Sensor Observation Service (SOS);
- Support for KML/KMZ Google Earth datasets;
- 3D buildings rendering of large urban areas based on the CityGML – The City Geography Markup Language;
- Planning and collaborative tools like distances and areas measurements, terrain profiling, moving objects tracking and advanced tools for what-if analysis;
- Massive 3D object rendering;
- OGC Compliance: Support for the open geospatial consortium standards: WMS, WFS, WFS-T, OpenLS, SOS, CityGML;
- Triangulated Irregular Network (TIN) generation and rendering support from multiresolution elevation;

The CesiumJS Middleware, on which it is based, is a client-side virtual globe and map library written in JavaScript using WebGL.

From an architectural point of view, it is organized and composed of four layers with increasing functionalities and abstraction:

- Core: the lowest layer, containing low-level widely used functions mostly related to math like linear algebra, intersection tests, and interpolation.
- Renderer: a thin abstraction over WebGL built-in GLSL uniforms, functions and abstractions for shader programs, textures and cube maps, buffers and vertex arrays, render states and frame buffers. Given vertex and fragment shader source strings, shader programs can be created directly from code invoking a function. Textures and cube maps have their abstraction in CesiumJS, invoking the binding texture functions of WebGL at higher level.
- Scene: works on top of Core and Renderer to provide high-level map and globe constructs like imagery layers, polylines, labels, cameras, ellipsoids, sensors and animations. It represents all the graphical objects and state for canvas.
- DynamicScene: enables data-driven visualization, primarily via the processing of CZML, a new JSON based schema for describing a time-dynamic graphical.

On top of these technologies, specific functionalities for the production planning within the mountainous areas have been developed:

Dynamic Visualization of digital surface model of the forest parcels as both regular and irregular grid of points: starting from the raster digital surface image a conversion in height map grid terrain files for the different level of detail is performed. Every terrain file represents one tile in the tiles quad-tree pyramid. The forest surface model dataset is served through a Tile Map Service, and is provided in the GeoBrowser3D as a height map surface for the virtual globe (figure 1). The surface model of the forest represents in this way the digital three dimensional model of the forest that replicates the real counterpart forest with a decimetre accuracy scale.

Terrain based distance and area measurements: within the planning system, users could enable analytical functionalities to measure distance between two points and polygonal area estimation inside the GeoBrowser3D interface. These tools enhance the capabilities of decisional support for the forestry planner and in particular in the phase of cableway planning design can be used as an accurate instrument for analysis of the constructional parameters within the forest model.

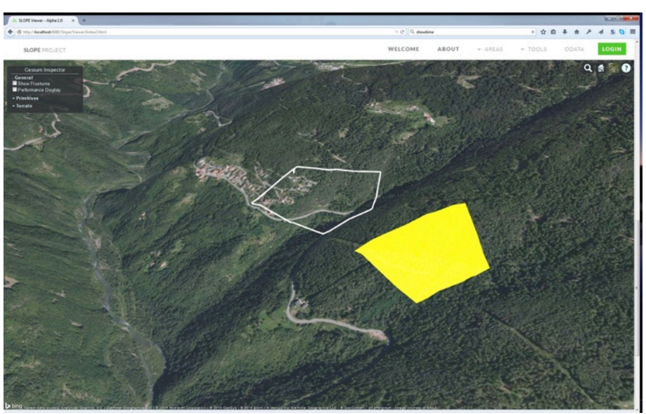

Figure 5: *Area and line measurement tools*.

High detail rendering of tree stems from on the field laser scans: laser field scan survey could be visualized as point cloud dataset. Lidar surveys results are available as point cloud LAS file where the space coordinates of each point can be retrieved and translate from the sensor reference system to a geographic one. This work developed a functionality to elaborate the laser scanner output LiDAR file and convert its coordinates to be compliant with the reference system used by CesiumJS middleware and have a compatible data structure to visualize every point as GeoBrowser3D geometry inside the virtual environment (figure 6).

Massive rendering of tree stem models of a forest derived from canopy analysis: the forest stems information is represented as a human readable JSON file, with tree plots information inferred from the analysis of laser scanner survey and UAV datasets. The input file is processed to represent and support the multiple visualization of stem characteristic of an entire forest parcel. In this way a virtual reproduction of the real counterpart forest is recreate with unique reproduction of each tree in its main physical and wood characteristic (figure 7).

Figure 6: *Point Cloud generated by UAV images matching visualized on the 3D environment*.

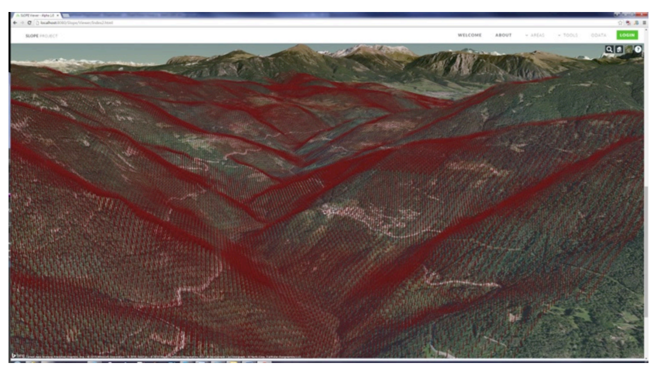

Figure 7: *Massive rendering of tree models on terrain*.

Real-time tracking of moving objects for transport monitoring and optimization: in the working area users could monitor in real time the position of the transport vehicles and the progress of harvesting and processing of the machines. The vehicle position is retrieved constantly, and represented as a three dimensional object in glTF format within the virtual globe with an animation showing its change of position. In the same way every machine will be represented as a model, which could be inspected to visualize information about quantity of wood processed, logs harvested and other operations performed on the stems, continuously querying from a database to update the data for work in progress.

Real-time placement of cable crane lines for harvesting in mountainous areas: with cable catenary computation, middle pillar placement, constraint checks and highlight of potential harvesting surface. More details about this feature are provided in the following section.

4.2 Computer aided cableway planning

Cableway planning in slope areas is often not trivial. Terrain height, forest density and accessibility take a role in the selection of pylons positions, height and number. Usually the setup of a cableway requires 2 days of work plus one day for each intermediate pylon that is needed. For this reason, minimizing the number of intermediate pylons is the priority to decrease timings and harvesting costs.

The remote data usually provides information for a whole forest parcel. In the Slope system, the so called technology layers define the possible operation areas for harvesting by cable yarder and excavator with the enhanced processor head. Depending on the forest management plan, the chosen area for harvesting can be selected by the forester from its PC, by zooming on the zone and load detailed terrain information. The planning starts by choosing position and height of pylons of the cableway. When the user enters position and height of a new pylon it gets connected to the previous one and the system check if the constraints are respected.

The steps taken by the system when the cableway is modified are:

1. Check for pylon constraints
2. Calculate cable function
3. Check for cable constraints
4. Calculate serviceable area.

The pylon and cable configurable constraints are:

- Length of the cable between two pylons: cables have limited length.
- Angle between previous and next pair of pylons: the cable can turn only a certain amount at each pylon.
- Height of the cable from the ground: the cable cannot be below a certain height from the terrain for proper functioning under load.
- Falling gradient: a minimum negative slope is required to pull the logs.

If all the constraints are respected the system places the new pylon and computes the area serviced by the cableway. The area is calculated based on cable height and terrain characteristics. From a technical point of view two lines at a certain angle are cast from each point on the cable and the points where the lines intersect the terrain are the limits of the area that can be serviced by the cableway. Points along the cable line with a low height will have a reduced harvesting area. This system provides a good yet simple estimation of the area serviceable by the cableway which is roughly assumed as two times the height of the cable line from the terrain in both directions.

The cable height evolution between pylons is computed following a catenary function, which describes the curve of a rope hanging from 2 points of a certain length supporting only his weight. Instead of using a fixed length cable based on the distance between the pylons, we calculated the required cable length to reach a certain fixed tensile force. This is what actually happens on the field, cables are tightened to a certain force, roughly 120KN. The catenary itself does not represent the position of the cable under the load of the carriage but it is a starting point from which we can estimate the cable behaviour during actual workload. To evaluate the validity of the selected pylon positions the system also calculates a maximum amount by which the cable can lower when under the load of the carriage.

The final cableway planning interface will let the user:

- Add additional pylons to the current cableway
- Move the pylons and recalculate the constraints
- Change the constraints parameters
- Save a cableway configuration
- Change tension and length of the cable

Pictures about the obtained results inside the Slope viewer are shown below.

5 Conclusions

In this study, it has been have presented the first results of the EU project SLOPE. The project aims are to develop a system allowing the planning and simulation of wood extraction from mountain forests. In particular the development of a forest visual simulation system combining technologies of 3D visualization and GIS through the web.

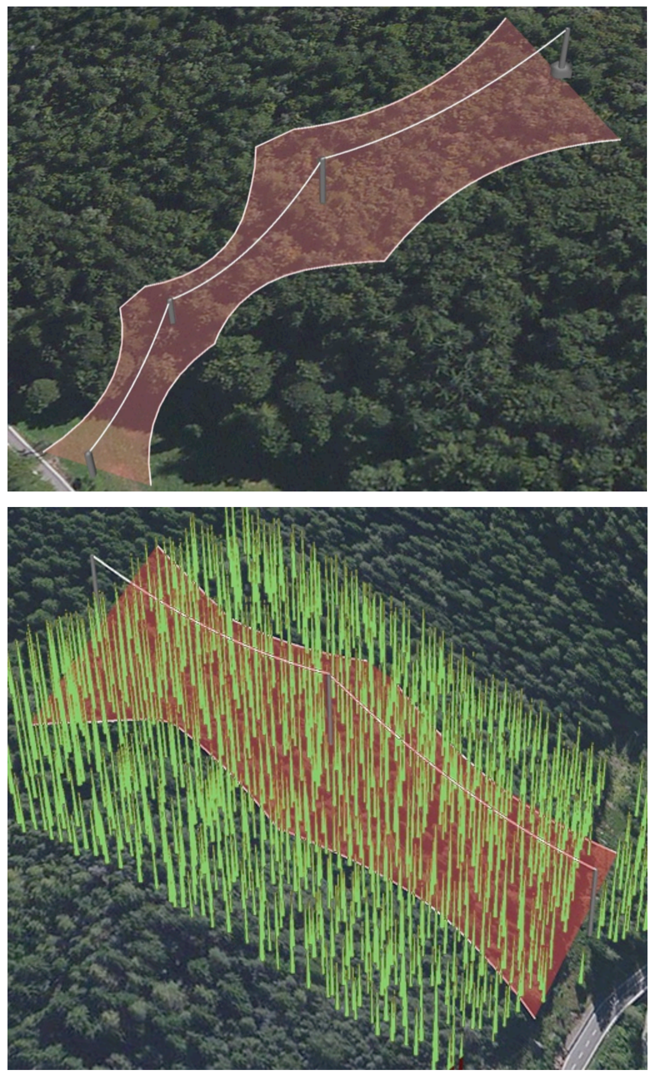

Figure 8: *Cableway placement and harvesting area simulation, with and without parcel tree stems visualization*.

The achievements obtained indicate that the combination of: standard services to provide information, service oriented architecture and the WebGL/HTML5 technologies make possible a 3D web based solutions to visualize, manage and simulate forestry data and operations.

The interactive visualization of the forest has been realized using a virtual globe representation that is, as already stated by [Simões and De Amicis, 2011], the ideal candidate to provide the most user-friendly experience in managing all the different needed information. Indeed, in our system, it is possible to combine different information coming from several data sources such as digital elevation produced by UAV survey, terrestrial laser data set and vector information available form map server.

At the highest level of detail, the simplified geometric models of individual trees are generated on the fly and faithful to underlying forest data stored in GIS systems. The hybrid representation method for 3D models and digital surface model can provide different representation based on the planning needs of the users. In addition, an implementation of a forest cable yard simulation module has been introduced and verified as being useful for constructing virtual environment for forest harvesting simulation in mountain areas and forest resources management.

The application presented within this document, is only the foundation of an integrated tool for forest production that will expose all the phases of the wood chain from tree marking to sawmill processing to a number of different actors.

After a finalization and optimization of the presented features, the system will be extended with ancillary features like cableway positioning by coordinate values and terrain height visualization along a path, the visualization system will be integrated into a new web user interface designed to assist all the actors of the forest production. This interface, shaped around forest actor's needs, will have the 3D visualization system as the main component and will allow the switch between three working modes: planning, deployment and monitoring. Each one provided with its own set of tools. This system will then be fully integrated with a forest information system acting as a repository of all the production and planning data, as well as the entry point for input data coming from UAV, TLS, Mobile devices and RFID tags. From a computer graphics point of view, additional research will be done to increase rendering performances of high detailed tree models and point cloud and to increase the effectiveness of shown tree properties (i.e. different colours for different stem qualities and values).

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